SunTax CH

Next-Generation AI Swiss Tax Platform • System Architecture & Client Demo Handover

Client Evaluation Release
Date: October 2026
Production Deployed

1. Live Cloud Endpoints

Component / Service	Live URL	Technical Details & Description
Frontend Web Application Vercel Production	**https://frontend-ruby-gamma-46.vercel.app/**	Next.js 14 App Router, TypeScript, responsive Swiss UI, live WebSocket updates.
Backend API Engine Render Cloud Service	**https://suntax-api.onrender.com**	FastAPI core with deterministic Swiss tax calculator, PyMuPDF OCR, PyOTP.
Interactive Swagger UI OpenAPI Documentation	**https://suntax-api.onrender.com/api/docs**	Interactive API testing sandbox for all authentication, document, and tax endpoints.
Technical Specification ReDoc Specification	**https://suntax-api.onrender.com/api/redoc**	Comprehensive OpenAPI 3.0 schema definitions, payloads, and error codes.
System Health Check	**https://suntax-api.onrender.com/health**	`Returns {'status': 'ok', 'environment': 'production'}.`

NOTE — Cloud Performance on Render Free Tier: The backend operates on Render free instance. If the service has been idle, the initial request awakens the container in ~45-60 seconds. All subsequent operations execute at sub-150ms latency.

2. GitHub Repository & Codebase Structure

Repository Detail	Configuration & URL	Architecture Notes
GitHub Repository	**https://github.com/usamashaikh13/suntax**	Monorepo with dedicated frontend/, backend/, and tax-rules/.
Production Branch	`main (Synced & Fully Tested)`	Continuous deployment triggered automatically on push to main.
Local Setup Helper	`./backend/start_dev.sh`	Automates venv creation, dependency installation, and server startup.

3. Database Architecture & Database File

Database Layer	File / Connection Target	Description & Security Policy
Development DB File	`backend/suntax_dev.db`	SQLite ACID database containing initialized user tables, tax returns, and profiles.
Production DB Engine	`PostgreSQL 16 (asyncpg)`	Asynchronous connection pool with Alembic schema migration management.
Data Isolation	`Row Level Security (RLS)`	User data strictly isolated; unauthorized cross-tenant queries blocked at DB layer.
Local Object Vault	`backend/dev_storage/`	Local encrypted file vault mirroring S3/MinIO bucket layout with user-prefixed keys.

4. Pre-Configured Demo Accounts

Account	Email & Password	2FA Security Status	Authenticator Setup / Code Access
Primary Demo (Full 2FA Active)	**client@suntax.ch** TestPassword123!	● **2FA Active** Required at Login & Final Submission	**Key:** OWZDANSQXHFBZXANGHT3ZKUYM2XW2AAV Or CLI: .venv/bin/python get_2fa.py client@suntax.ch
Secondary Demo (Standard Access)	**demo@suntax.ch** TestPassword123!	■ **Direct Login** Ready for live 2FA setup demo	Can be used to demonstrate live 2FA QR code pairing in *My Account* → *Security*.

5. Step-by-Step Client Testing Guide (5-Minute Verification)

Step 1: Authenticate	Navigate to /login on Vercel or localhost. Log in with **client@suntax.ch**. Enter the 6-digit TOTP code from your mobile authenticator app (Google Authenticator / Apple Passwords).
Step 2: Create Declaration	Click **'+ New Tax Return'**. Select Canton **ZH (Zürich)**, Municipality **Zürich (BFS #261)**, and Tax Year **2025**. Click Confirm.
Step 3: Upload Documents	In the **Documents** tab, drag & drop the test documents (Salary Certificate, UBS Bank Statement, Pillar 3a Certificate, Donation Receipt).
Step 4: Review OCR Data	Click **'Review Fields'** on any document marked Needs Review. Review the extracted figures and click **'Apply to Tax Profile'**.
Step 5: Compute Taxes	Switch to the **Calculation** tab and click **'Calculate Tax'**. Deterministic calculation resolves: Federal CHF 637, Canton CHF 1'852.40, Municipal CHF 2'204.35 (Total: CHF 4'693.75).
Step 6: Step-Up 2FA Submit	Under **Review & Submit**, check the Art. 110 DBG declaration box. Click **'Confirm Tax Return Draft'**. Enter your 6-digit 2FA security code to seal and lock the filing.

6. Key API Endpoints Reference

Method & Path	Endpoint Category	Operational Description
`POST /api/v1/auth/login`	Authentication	Validates email & password. If 2FA enabled, issues temporary token for TOTP challenge.
`POST /api/v1/auth/login/2fa`	2FA Challenge	Exchanges temporary token + 6-digit TOTP code for full-scope JWT access token.
`GET /api/v1/tax-returns`	Tax Filings	Retrieves paginated list of tax declarations owned by authenticated user.
`POST /api/v1/tax-returns/{id}/calculate`	Deterministic Engine	Applies verified cantonal and federal tax brackets to user's TaxProfile.
`POST /api/v1/tax-returns/{id}/confirm`	Legal Confirmation	Seals declaration under Art. 110 DBG; validates 6-digit 2FA code if TOTP enabled.
`POST /api/v1/tax-returns/{id}/export/pdf`	Official Export	Generates complete official Swiss tax declaration summary PDF.